

# TidyTuesday Week 39

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Tuesday 22<sup>nd</sup> September, 2026

## Contents

|          |                                |          |
|----------|--------------------------------|----------|
| <b>1</b> | <b>Weekly Summary</b>          | <b>2</b> |
| <b>2</b> | <b>Date: 2026-09-21</b>        | <b>3</b> |
| 2.1      | Raw Regression . . . . .       | 3        |
| 2.2      | Detrended Regression . . . . . | 5        |
| <b>3</b> | <b>Date: 2026-09-22</b>        | <b>7</b> |
| 3.1      | Raw Regression . . . . .       | 7        |
| 3.2      | Detrended Regression . . . . . | 9        |

# 1 Weekly Summary

## AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to another thrilling week of shoving unrelated economic data into a blender and pressing "regress." As always, we are drawing zero real conclusions from any of this, because the whole enterprise is a monument to statistical mischief. But my goodness, did the numbers put on a show.

Monday's pairing was a real work of art: New England commercial electric power prices versus a national price index for "Symptoms, Signs, and Ill-Defined Conditions" medical spending. Yes, that is a real series, and no, I did not make up that name to sound spooky. The raw regression came roaring out of the gate with an R-squared of 0.93, which is the kind of number that makes a person want to call a press conference and announce that New England electricity secretly governs the cost of America's vaguest ailments. And then we detrended it, and the whole thing evaporated into thin air – R-squared of 0.02, a p-value of basically 0.51, a coefficient so close to zero it needs a magnifying glass. Textbook trend mirage. Two series wandered upward over the decades holding hands, we assumed romance, and detrending revealed they were simply two strangers escalating the same staircase. Absolutely nothing to see here, which is exactly what we love to see.

Tuesday, however, got weird in the fun way. We matched the active real estate listing count in Flathead County, Montana – a specific and delightful place – against U.S. imports of maintenance and repair services. The raw regression was actually the weaker of the two here (R-squared a humble 0.12), and then, in a plot twist worthy of a dramatic gasp, the detrended version got stronger, climbing to 0.19 with a p-value hovering around four in ten million. That means once you strip out the long-run drift from both, there's still a wiggle-for-wiggle relationship left standing. In this project, a detrended survivor is the rarest of beasts, so allow me to over-emote appropriately: incredible, unprecedented, someone alert Montana. My extremely scientific theory is that when houses pile up on the market in Flathead County, the nation collectively decides it's time to fix things, and imported repair services surge. Is that true? Almost certainly not. Is it a fun sentence? Yes.

So this week gave us one perfect fake and one suspiciously durable coincidence, which is a fantastic ratio for a project whose entire premise is that spurious correlation is the norm and real relationships are the exception. Please do not refinance anything, relocate to Flathead County, or short New England electricity based on these results. We are two random series and a linear model out here, having a laugh. See you next week for more numbers doing things they were never asked to do.

## 2 Date: 2026-09-21

### Series ID: PCU221122221122421

This series is titled Producer Price Index by Industry: Electric Power Distribution: Commercial Electric Power for New England Census Division and has a frequency of Monthly. The units are Index Dec 1990=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1971-01-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

### Series ID: SYSIILPIBLEND

This series is titled Medical Services Expenditures by Disease: Symptoms; Signs; and Ill-Defined Conditions Price Index, Blended Account Basis and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2021-01-01. The popularity of this series is 1.

## 2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

|                          |                          |                            |          |
|--------------------------|--------------------------|----------------------------|----------|
| <b>Dep. Variable:</b>    | value_fred_SYSIILPIBLEND | <b>R-squared:</b>          | 0.933    |
| <b>Model:</b>            | OLS                      | <b>Adj. R-squared:</b>     | 0.929    |
| <b>Method:</b>           | Least Squares            | <b>F-statistic:</b>        | 277.7    |
| <b>Date:</b>             | Mon, 21 Sep 2026         | <b>Prob (F-statistic):</b> | 3.41e-13 |
| <b>Time:</b>             | 17:16:08                 | <b>Log-Likelihood:</b>     | -68.300  |
| <b>No. Observations:</b> | 22                       | <b>AIC:</b>                | 140.6    |
| <b>Df Residuals:</b>     | 20                       | <b>BIC:</b>                | 142.8    |
| <b>Df Model:</b>         | 1                        |                            |          |
| <b>Covariance Type:</b>  | nonrobust                |                            |          |

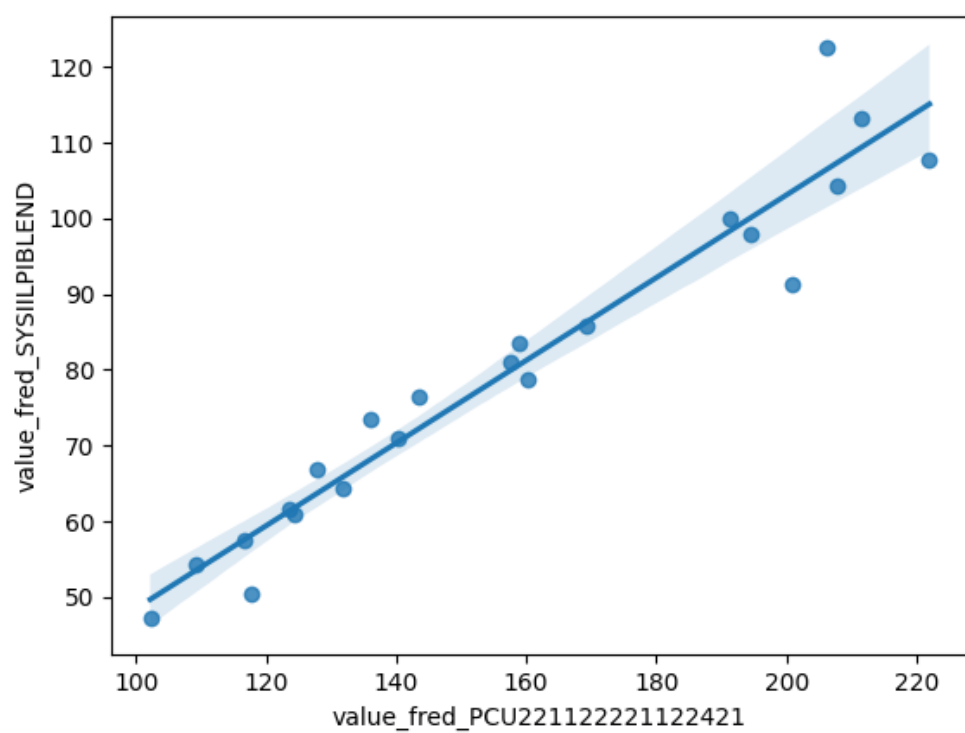
|                                      | coef    | std err | t      | P>  t | [0.025  | 0.975] |
|--------------------------------------|---------|---------|--------|-------|---------|--------|
| <b>const</b>                         | -6.0876 | 5.281   | -1.153 | 0.263 | -17.103 | 4.928  |
| <b>value_fred_PCU221122221122421</b> | 0.5458  | 0.033   | 16.664 | 0.000 | 0.477   | 0.614  |

|                       |       |                          |        |
|-----------------------|-------|--------------------------|--------|
| <b>Omnibus:</b>       | 6.881 | <b>Durbin-Watson:</b>    | 1.268  |
| <b>Prob(Omnibus):</b> | 0.032 | <b>Jarque-Bera (JB):</b> | 5.911  |
| <b>Skew:</b>          | 0.534 | <b>Prob(JB):</b>         | 0.0521 |
| <b>Kurtosis:</b>      | 5.304 | <b>Cond. No.</b>         | 706.   |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 1:** Raw Regression Plot for 2026-09-21

## 2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

|                          |                                    |                            |         |
|--------------------------|------------------------------------|----------------------------|---------|
| <b>Dep. Variable:</b>    | value_fred_SYSIILPIBLEND_detrended | <b>R-squared:</b>          | 0.022   |
| <b>Model:</b>            | OLS                                | <b>Adj. R-squared:</b>     | -0.027  |
| <b>Method:</b>           | Least Squares                      | <b>F-statistic:</b>        | 0.4428  |
| <b>Date:</b>             | Mon, 21 Sep 2026                   | <b>Prob (F-statistic):</b> | 0.513   |
| <b>Time:</b>             | 17:16:08                           | <b>Log-Likelihood:</b>     | -54.441 |
| <b>No. Observations:</b> | 22                                 | <b>AIC:</b>                | 112.9   |
| <b>Df Residuals:</b>     | 20                                 | <b>BIC:</b>                | 115.1   |
| <b>Df Model:</b>         | 1                                  |                            |         |
| <b>Covariance Type:</b>  | nonrobust                          |                            |         |

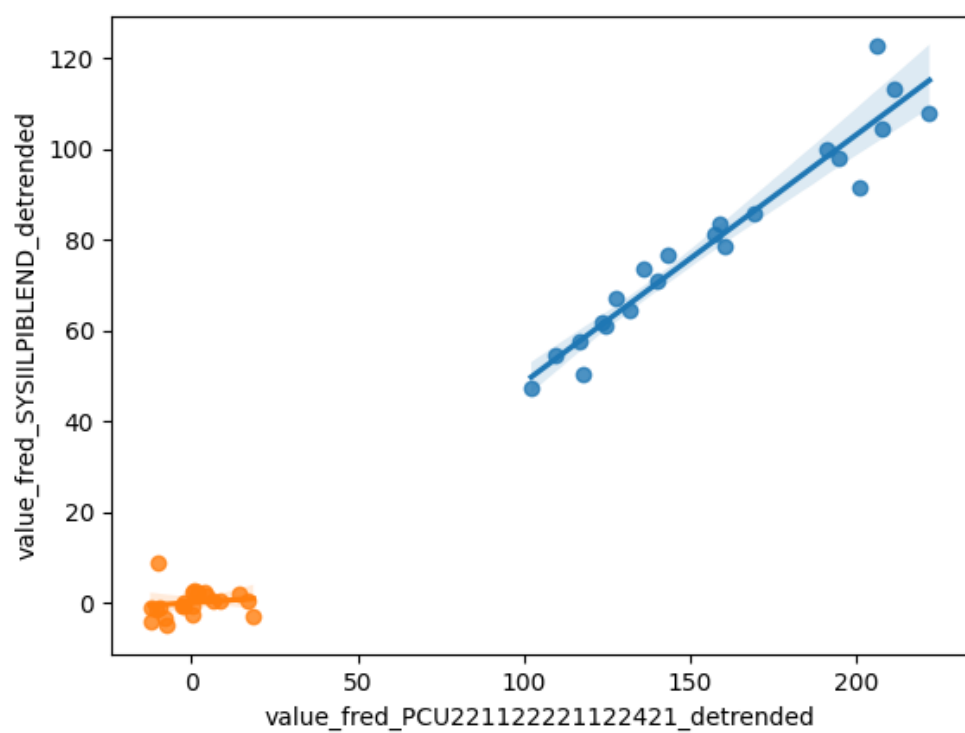
|                                                | coef      | std err | t        | P>  t | [0.025 | 0.975] |
|------------------------------------------------|-----------|---------|----------|-------|--------|--------|
| <b>const</b>                                   | 4.119e-14 | 0.643   | 6.41e-14 | 1.000 | -1.340 | 1.340  |
| <b>value_fred_PCU221122221122421_detrended</b> | 0.0480    | 0.072   | 0.665    | 0.513 | -0.103 | 0.199  |

|                       |        |                          |         |
|-----------------------|--------|--------------------------|---------|
| <b>Omnibus:</b>       | 12.738 | <b>Durbin-Watson:</b>    | 0.457   |
| <b>Prob(Omnibus):</b> | 0.002  | <b>Jarque-Bera (JB):</b> | 12.678  |
| <b>Skew:</b>          | 1.200  | <b>Prob(JB):</b>         | 0.00177 |
| <b>Kurtosis:</b>      | 5.841  | <b>Cond. No.</b>         | 8.90    |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 2:** Detrended Regression Plot for 2026-09-21

### 3 Date: 2026-09-22

#### Series ID: ACTLISCOU30029

This series is titled Housing Inventory: Active Listing Count in Flathead County, MT and has a frequency of Monthly. The units are Level and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2016-07-01 and the observation end date is 2026-08-01. The popularity of this series is 2.

#### Series ID: ITMMARM133S

This series is titled U.S. Imports of Services: Maintenance and Repair Services, not included elsewhere and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1999-01-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

### 3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

|                          |                        |                            |          |
|--------------------------|------------------------|----------------------------|----------|
| <b>Dep. Variable:</b>    | value_fred_ITMMARM133S | <b>R-squared:</b>          | 0.123    |
| <b>Model:</b>            | OLS                    | <b>Adj. R-squared:</b>     | 0.116    |
| <b>Method:</b>           | Least Squares          | <b>F-statistic:</b>        | 16.76    |
| <b>Date:</b>             | Tue, 22 Sep 2026       | <b>Prob (F-statistic):</b> | 7.76e-05 |
| <b>Time:</b>             | 15:49:46               | <b>Log-Likelihood:</b>     | -747.35  |
| <b>No. Observations:</b> | 121                    | <b>AIC:</b>                | 1499.    |
| <b>Df Residuals:</b>     | 119                    | <b>BIC:</b>                | 1504.    |
| <b>Df Model:</b>         | 1                      |                            |          |
| <b>Covariance Type:</b>  | nonrobust              |                            |          |

|                           | coef     | std err | t      | P>  t | [0.025  | 0.975]  |
|---------------------------|----------|---------|--------|-------|---------|---------|
| const                     | 452.9830 | 36.286  | 12.484 | 0.000 | 381.134 | 524.832 |
| value_fred_ACTLISCOU30029 | 0.1837   | 0.045   | 4.094  | 0.000 | 0.095   | 0.273   |

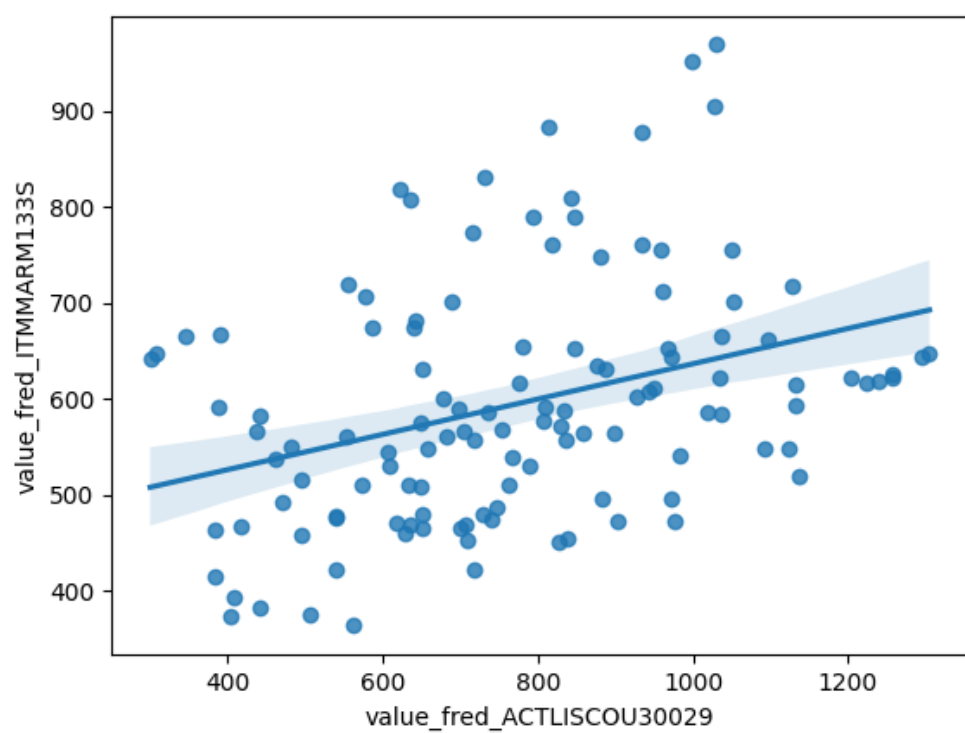
  

|                       |        |                          |          |
|-----------------------|--------|--------------------------|----------|
| <b>Omnibus:</b>       | 12.594 | <b>Durbin-Watson:</b>    | 0.214    |
| <b>Prob(Omnibus):</b> | 0.002  | <b>Jarque-Bera (JB):</b> | 14.022   |
| <b>Skew:</b>          | 0.833  | <b>Prob(JB):</b>         | 0.000902 |
| <b>Kurtosis:</b>      | 3.056  | <b>Cond. No.</b>         | 2.75e+03 |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.75e+03. This might indicate that there are strong multicollinearity or other numerical problems.



**Figure 3:** Raw Regression Plot for 2026-09-22

### 3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

|                          |                                  |                            |          |
|--------------------------|----------------------------------|----------------------------|----------|
| <b>Dep. Variable:</b>    | value_fred_ITMMARM133S_detrended | <b>R-squared:</b>          | 0.195    |
| <b>Model:</b>            | OLS                              | <b>Adj. R-squared:</b>     | 0.188    |
| <b>Method:</b>           | Least Squares                    | <b>F-statistic:</b>        | 28.77    |
| <b>Date:</b>             | Tue, 22 Sep 2026                 | <b>Prob (F-statistic):</b> | 4.07e-07 |
| <b>Time:</b>             | 15:49:46                         | <b>Log-Likelihood:</b>     | -740.70  |
| <b>No. Observations:</b> | 121                              | <b>AIC:</b>                | 1485.    |
| <b>Df Residuals:</b>     | 119                              | <b>BIC:</b>                | 1491.    |
| <b>Df Model:</b>         | 1                                |                            |          |
| <b>Covariance Type:</b>  | nonrobust                        |                            |          |

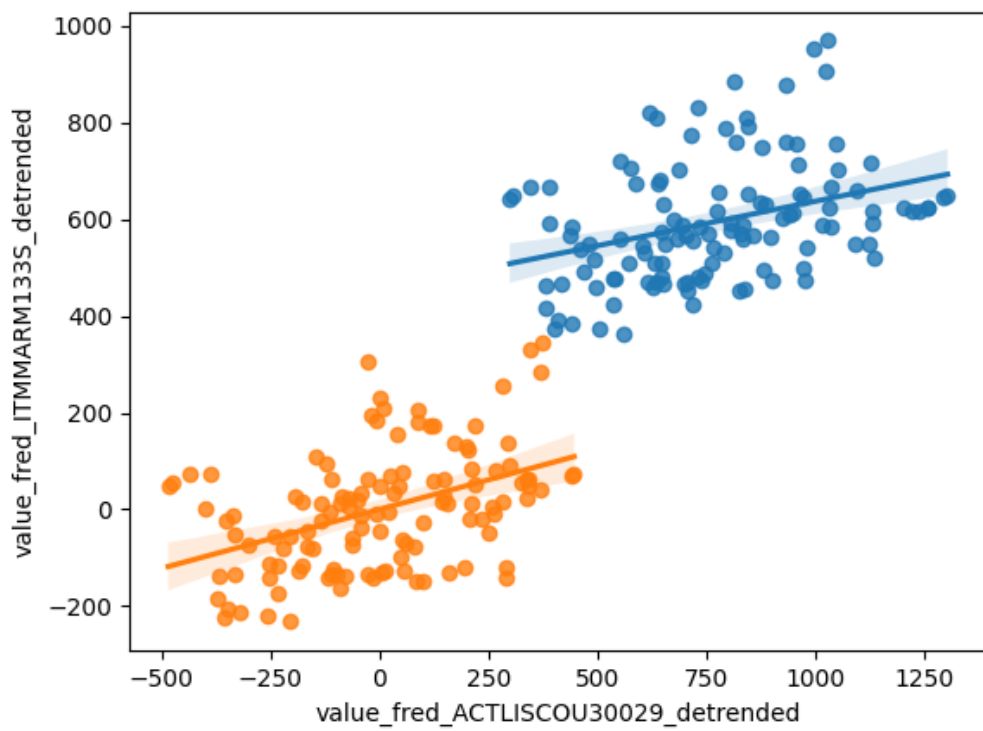
|                                            | coef       | std err | t         | P>  t | [0.025  | 0.975] |
|--------------------------------------------|------------|---------|-----------|-------|---------|--------|
| <b>const</b>                               | -3.037e-11 | 10.104  | -3.01e-12 | 1.000 | -20.008 | 20.008 |
| <b>value_fred_ACTLISCOU30029_detrended</b> | 0.2437     | 0.045   | 5.364     | 0.000 | 0.154   | 0.334  |

|                       |       |                          |        |
|-----------------------|-------|--------------------------|--------|
| <b>Omnibus:</b>       | 4.932 | <b>Durbin-Watson:</b>    | 0.254  |
| <b>Prob(Omnibus):</b> | 0.085 | <b>Jarque-Bera (JB):</b> | 5.006  |
| <b>Skew:</b>          | 0.473 | <b>Prob(JB):</b>         | 0.0819 |
| <b>Kurtosis:</b>      | 2.689 | <b>Cond. No.</b>         | 222.   |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 4:** Detrended Regression Plot for 2026-09-22